

Asutosh Dalei

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WORK EXPERIENCE

Machine Learning Engineering Intern

06/2025 – 06/2026

Nokia – Sunnyvale, California

▪ Knowledge Layer & RAG Architecture:

- Designed and built the “**Knowledge Layer**”, a retrieval engine for querying complex technical documentation by unifying Vector Search (OpenSearch), relational metadata (PostgreSQL), and graph traversal (Neo4j).
- Benchmarked 4+ SOTA embedding models against a **validation corpus of 700+ technical documents**. Selected **Embedding Gemma** for the best MRR and semantic recall. Built a custom sentence-aware chunking pipeline to optimize ingestion.
- Built a **Multimodal RAG pipeline** to index unstructured tables and figures using a joint vector space with **Jina Embeddings V4** and **OpenAI CLIP**, enabling text-to-visual retrieval.
- Added a re-ranking stage with **Qwen3 Reranker-8B** (cross-encoder), measurably improving context precision in retrieved chunks.
- Built an agentic workflow with the **AG2 framework** to automate multi-step information gathering across the knowledge base.

Stack: Docker, Kafka, PostgreSQL, OpenSearch, Neo4j, MinIO, FastAPI, Python, GCP, AG2, vLLM.

Data Scientist

01/2022 – 08/2024

Maruti Suzuki (R&D Division) – Bengaluru, India

▪ Battery Health Prediction (Predictive Maintenance):

- Built an **LSTM-based** time-series classifier on telematics data to predict lead-acid battery failure in connected vehicles. Trained on imbalanced sensor data using **focal cross-entropy loss**.
- Designed a **14-day sliding window** aggregation layer over raw predictions to suppress sensor noise and cut false positives before alerts reach the field.
- Deployed across **100K+ connected vehicles** with **98% field-validated accuracy**, reducing warranty claims and roadside failures. Work resulted in a published patent (No. 202411040338).

▪ MLOps & Training Cost Reduction:

- Audited a legacy training pipeline burning **\$4,000/month** in GPU compute. Profiled bottlenecks and migrated Random Forest workloads to optimized multi-core CPU execution, eliminating unnecessary GPU spend.
- Integrated **Optuna** for automated hyperparameter search, cutting per-experiment wall time and total cloud compute costs.

EDUCATION

Master of Science, Data Science

GPA: 3.97 / 4.0

University of Maryland, College Park

08/2024 – 05/2026

- Relevant Coursework: Advanced Machine Learning, Natural Language Processing, Probability & Statistics, Data Representation & Modelling, Big Data Systems, Cloud Computing.

Bachelor of Technology, Electrical and Electronics Engineering

GPA: 9.1 / 10

Vellore Institute of Technology, Vellore

06/2018 – 06/2022

RESEARCH & PATENTS

Patent – Automobile Battery Health Prediction System (No. 202411040338)

- **Primary ML contributor** on a patented system for predicting remaining useful life of lead-acid batteries using telematics sensor data and LSTM-based deep learning. Now deployed in production across **100K+** Maruti Suzuki’s connected vehicle fleet.

Molecular Signatures and ML-driven Stress Biomarkers for Rainbow Trout Aquaculture – *Nature Scientific Reports*, 2025

- Built **predictive models on genomic data** to identify stress biomarkers in rainbow trout. Contributed model design, feature selection across high-dimensional gene expression datasets, and analysis of markers linked to environmental stress response.

TECHNICAL SKILLS

- **ML & AI:** PyTorch, TensorFlow, HuggingFace, Transformers, LLM Fine-Tuning (SFT, RLHF, PEFT), RAG, Agentic AI, LangChain, LLM Evaluation, Structured Output Enforcement, Inference Optimization, Model Serving (vLLM, SGLang).
- **Experiment Tracking & MLOps:** MLflow, Optuna, Weights & Biases, GitHub Actions.
- **Languages:** Python, GoLang, R, SQL, MATLAB, C, C++.
- **Distributed & Data Systems:** Kafka, FastAPI, Docker, Kubernetes.
- **Cloud & Infrastructure:** GCP, Azure, Linux, Git.
- **Databases:** PostgreSQL, MongoDB, MySQL, OpenSearch, Neo4j.

PROJECTS

Agentic RAG over Enterprise Knowledge Graphs

- Designed a enterprise-ready modular agentic framework (**AG2**) for natural-language querying over enterprise knowledge graphs spanning **Ab Initio** workflows. Deployed on **Kubernetes**, served via **Vertex AI**, with **ZeRank2** re-ranking for retrieval precision.

Spatiotemporal Modeling of Telematics Data Gaps

- Solved recurring multi-hour data dropouts (5% of telematics records) to regional network dead zones. Built an ML pipeline using Random Forests for anomaly classification and **GRU over Spatiotemporal Sequences** to map affected geographic clusters.
- Improved data collection completeness by **15%** across connected vehicle fleet.